



Customer Shopping Behavior Analysis

Uncovering Insights from 3,900 Customer Transactions

Project Overview



Transactional Data

Analyzes 3,900 purchases across various product categories to uncover behavioral patterns.



Key Focus Areas

Spending patterns, customer segments, product preferences, and subscription behavior.



Business Goal

Guide strategic decisions with data-driven insights from end-to-end analysis.



Dataset Summary

3,900

Rows

Customer transactions

18

Columns

Key features

37

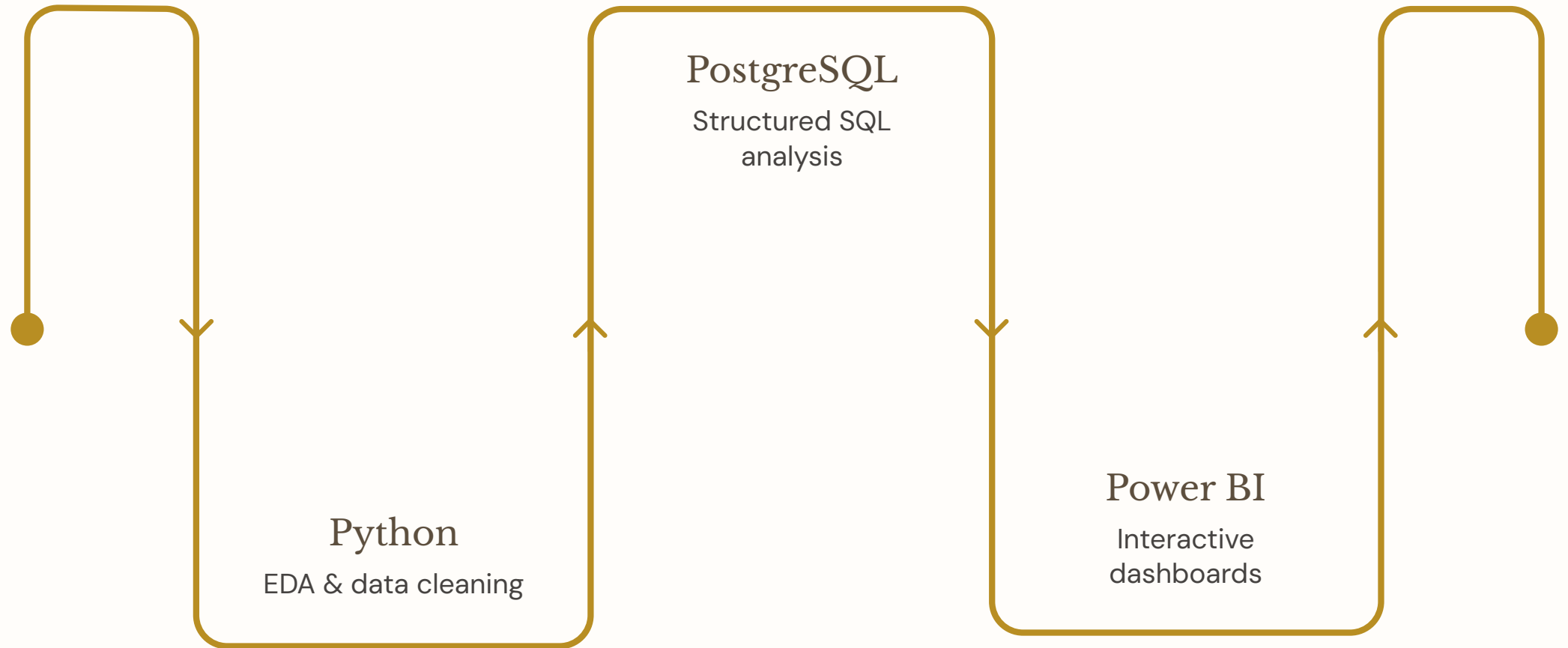
Missing Values

Review Rating — handled via median imputation

 **Key Features:** Customer Demographics, Purchase Details, Shopping Behavior

 **Data Quality:** 37 missing values in Review Rating column resolved through category-based median imputation.

Tools & Workflow



An end-to-end data pipeline moving from exploratory analysis and cleaning in Python, through structured querying in PostgreSQL, to interactive visualization in Power BI.

Exploratory Data Analysis in Python

Data Loading & Exploration

Initial inspection of structure, distributions, and data types.

Missing Value Handling

Median imputation applied by category for Review Rating.

Column Standardization

All column names converted to snake_case for consistency.

Feature Engineering

Created `age_group` and `purchase_frequency_days` derived columns.

Data Consistency

Dropped redundant `promo_code_used` column.

Database Integration

Cleaned dataset loaded into PostgreSQL for SQL analysis.

10 Business Questions Answered via SQL

- Revenue by Gender
- High-Spending Discount Users
- Top 5 Products by Rating
- Shipping Type Comparison
- Subscribers vs Non-Subscribers
- Discount-Dependent Products
- Customer Segmentation — New, Returning, Loyal
- Top 3 Products per Category
- Repeat Buyers & Subscriptions
- Revenue by Age Group

Customer Segmentation

Customers are segmented by purchase history to enable targeted engagement strategies.

 New

Less than 2 previous purchases

 Returning

2 to 10 previous purchases

 Loyal

More than 10 previous purchases



Interactive Power BI Dashboard



Revenue by Category

Visual breakdown of revenue across product categories.



Subscription Status

% of customers by subscription status with slicers for gender, category, and subscription.



Shipping Type Filter

Interactive filter to analyze sales by shipping type.



Sales by Category

Category-level sales performance with drill-down capability.



Business Recommendations



Boost Subscriptions

Promote exclusive subscriber benefits to increase conversion.



Loyalty Programs

Reward repeat buyers to strengthen retention.



Review Discount Policy

Balance sales growth with margin control.



Product Positioning

Highlight top-rated and best-selling products.



Targeted Marketing

Focus on high-revenue age groups and express shipping users.



Conclusion

This analysis delivered actionable insights across customer demographics, purchase behavior, and product performance — using an end-to-end data pipeline from **Python** to **SQL** to **Power BI**.

Python

EDA & Cleaning

PostgreSQL

SQL Analysis

Power BI

Dashboard